

Federica Domecq Lacroze

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EDUCATION

Pomona College

Double Major: Mathematics & Computer Science

GPA: 3.97/4.00

Claremont, CA

Aug. 2022 – May 2026

Relevant coursework:

- **Computer Science:** Data Visualization (Fall 2025); Machine Learning; Algorithms; Data Structures & Algorithms with Java; Languages & Theory; Functional Programming & Discrete Math
 - **Mathematics:** Computational Statistics (Fall 2025); Statistical Theory; Time Series; Deterministic Operations Research; Real Analysis I & II; Probability; Combinatorics; Vector Calculus; Linear Algebra; Statistics with R
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EXPERIENCE

Pomona College

Claremont, CA

Kenneth Cooke Independent Research Fellow – Strategic Golf Powered by ML

May 2025 – Aug. 2025

- Modeled expected strokes to hole out using Gaussian Process Regression over rasterized satellite hole layouts, producing smooth outcome surfaces for shot distributions.
- Integrated TrackMan shot data to personalize aimpoint and club selection, applying custom loss functions to reflect player goals (e.g., birdie-seeking, hazard avoidance).
- Built a sequential decision framework for full-hole play, incorporating spatial risk and player tendencies to simulate optimal tee-to-green strategies.

Summer Research Assistant, Department of Mathematics

May 2024 – Aug. 2024

- Validated the **sicegar** R package for detecting gene expression onset times by generating synthetic datasets and computing numerical inflection points for nonlinear models.
- Designed large-scale simulations that varied data density, sparsity, sampling rate, and variability, parallelizing workflows on high-performance computing (HPC) clusters.
- Benchmarked **sicegar** against alternative tools (**ImpulseDE2**, **DESeq2**) and developed a bootstrap-based correction that improved detection accuracy, contributing to a conference talk and manuscript.

Teaching Assistant – Statistics, Data Science & Probability

Jan. 2024 – Present

- Lead tri-weekly sessions on R programming, supporting students with hypothesis testing, confidence intervals, t-tests, chi-square tests, ANOVA, regression, and residual analysis.
- Guided students on data wrangling, simulation, and visualization using **tidyverse** packages.

Resident Advisor

Dec. 2024 – May 2025

- Supervised 100+ students in a residential area of 400, fostering an inclusive environment, resolving conflicts, and managing crises.
- Planned 5+ community programs per semester, enforced college policies, documented incidents, and liaised with campus staff on student welfare.

DoctorSIM

Madrid, Spain

Marketing & Administrative Intern

Dec. 2021 – Jul. 2022

- Built a market research database of 50+ potential clients for backlinking, improving search engine optimization.
- Managed invoicing and coordinated thousands of DHL shipments remotely.

Summer Intern

Summer 2019

- Maintained internal product databases, created content to improve Google search ranking, and processed hundreds of financial transactions with fraud risk.
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ACTIVITIES & LEADERSHIP

Women's Varsity Golf – Team Captain

Pomona-Pitzer, 2022 – 2026

- Led team to 3rd-place finish at NCAA Championship; earned 2 collegiate titles; 2× Top-5 finishes at Conference Championships; 2× All-Conference & All-Region Awards (2024, 2025); WGCA All-American Scholar (2023–2025).

Co-President – Association for Women in Mathematics (Claremont)

2023 – 2025

Math Department Liaison

2025 – Present

SKILLS & LANGUAGES

Programming: Python (Pandas); R (Tidyverse); JavaScript (D3); Haskell; $\text{\LaTeX}$; HTML/CSS

Tools: Git; Excel; High-Performance Computing & Parallel Processing; LLM-assisted coding

Languages: English (Fluent); Spanish (Fluent); French (Intermediate); Portuguese (Intermediate)